

Topic 6B: Alkanes
8. know the general formula for alkanes
9. know that alkanes and cycloalkanes are saturated hydrocarbons
10. understand that alkane fuels are obtained from the fractional distillation, cracking and reforming of crude oil <i>Reforming is described as the processing of straight-chain hydrocarbons into branched-chain alkanes and cyclic hydrocarbons for efficient combustion.</i>
11. know that pollutants, including carbon monoxide, oxides of nitrogen and sulfur, carbon particulates and unburned hydrocarbons, are formed during the combustion of alkane fuels

Students should:

- | |
|---|
| 12. understand the problems arising from pollutants from the combustion of fuels, limited to the toxicity of carbon monoxide and the acidity of oxides of nitrogen and sulfur |
| 13. understand how the use of a catalytic converter solves some problems caused by pollutants |
| 14. understand the use of alternative fuels, including biodiesel and alcohols derived from renewable sources such as plants, in terms of a comparison with non-renewable fossil fuels |
| 15. know that a radical: <ul style="list-style-type: none">i is a species with an unpaired electron and is represented in mechanisms by a single dotii is formed by homolytic fission of a covalent bond and results in the formation of radicals |
| 16. understand the reactions of alkanes with: <ul style="list-style-type: none">i oxygen in air (combustion)ii halogens, in terms of the mechanism of radical substitution through initiation, propagation and termination steps <p><i>The use of curly half-arrows is not expected in this mechanism.</i></p> |
| 17. understand the limitations of the use of radical substitution reactions in the synthesis of organic molecules, in terms of further substitution reactions and the formation of a mixture of products |